Jackson Huang

CA/US Citizen | 647-928-5526 | j672huan@uwaterloo.ca | jacksonhuang.ca

Education

University of Waterloo Sep 2025 – Present

Bachelor of Applied Science in Nanotechnology Engineering

Experience

The Maverik July 2026 – Aug 2026

Data Analyst

- Delivered **9** structured business deliverables for Maverik, a Canadian women-owned cleaning brand, including surface care category analysis, consumer segmentation, competitive positioning, and market sizing
- Identified Scouring Agents as the fastest-growing category segment at **13.2% CAGR** through **2028** and pinpointed Safix as the client's closest true competitor through primary and secondary market research
- Built a multi-page interactive Power BI dashboard with **5+ visualizations** analyzing CO_2 emissions per capita across **150+ countries** for a UN SDG 13 (Climate Action) project, identifying and correcting a DAX logic error (SUM vs. weighted average)
- Coordinated as question-gatherer for weekly stakeholder meetings across a team of **8**, consolidating **15+** team questions ahead of client check-ins

Radiant Exhibits Feb 2024 – June 2024

SEO Specialist

- Optimized on-page SEO for **25+** client webpages, contributing to an average **15–20%** increase in organic search traffic over one quarter
- Conducted keyword research and competitor analysis across **3** major competitors, identifying **40+** high-impact keywords used to guide content strategy
- Designed and optimized technical layout diagrams for medical blood bag systems, balancing spatial constraints with display clarity for client-facing product documentation

Projects

Triboelectric Nanogenerator (TENG) | *MEMS, Energy Harvesting*

- Designed and built a MEMS-based triboelectric nanogenerator harvesting energy from airflow-induced motion at frequencies up to **10 Hz**
- Implemented a contact–separation mechanism generating peak voltages on the order of **tens of volts**

TalkRetro | *React, Vite, Node.js, Express, Socket.io, PostgreSQL, Redis*

- Built a real-time chat app with a custom Windows XP-themed UI, supporting DMs and group chats with replies, reactions, and media sharing for **50+** test users
- Used Socket.io and Redis to power live presence and typing indicators across **15+** concurrent conversations, with PostgreSQL for persistent message storage
- Integrated Clerk for authentication and resolved **10+** bugs, including user-picker and invite/leave logic errors, improving delivery reliability to **99%**

Materials Property Explorer | *JavaScript, Node.js, HTML/CSS*

- Developed a full-stack web application for querying and visualizing materials property data via the Materials Project API, covering **10+** material properties
- Built a Node.js backend server to handle **100+** API requests and serve structured materials data to a dynamic frontend interface

Technical Skills

Languages: Python, SQL, JavaScript, Java, HTML, CSS, MatLab
Technologies: AutoCAD, Solidworks, Oscilloscope, Multimeter, DMM, Breadboarding, React, Node.js, Express, Vite, Firebase, Firebase Authentication, Clerk, Socket.io, PostgreSQL, Redis, Git, GitHub, VS Code, PowerBI, DAX, Google Analytics, SEMrush, Ahrefs, MS Office, Wordpress
Skills: Energy harvesting, Microfabrication, Data visualization, Dashboard development, Database design, User authentication, Protected routes, Real-time communication systems, Technical troubleshooting